

Ashwin Krishnamurthy

Mechatronics engineer building the future of space and defense tech.

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EXPERIENCE

Materials Science Research Intern Dec 2024 – Present
Rutgers University microMechanics of Deformation Group *Piscataway, NJ*

- Automated disconnection extraction using Interfacial Line Defect Analysis and custom Python scripts.
- Computed defect content for 7,304 previously unstudied energy-minimized aluminum structures on HPC, replacing one-by-one processing.
- Preparing a publication on the relationship between disconnections, grain boundary misorientation, and migration.

Astronomy Research Intern May 2025 – Dec 2025
NASA STEM Enhancement in Earth Science (SEES) *Remote*

- Refined distance and age calculations for open cluster NGC 2194; presented at the American Geophysical Union and MIT.
- Performed photometric analysis in AstrolmageJ on 700+ previously unidentified stars to construct a color-magnitude diagram.

Observatory Intern May 2024 – Present
New Jersey Astronomical Association *Glen Gardner, NJ*

- Certified operator of the observatory's 26-inch telescope for public programs and astrophotography.
- Captured 100+ images of deep-sky objects for a personal portfolio; led star party booths, recruiting 80+ new members.

PROJECTS & LEADERSHIP

NASA App Development Challenge, Co-Captain Sep 2024 – May 2025

- National winner for a simulation of the Artemis II mission with key flight data and an optimized communications schedule.
- Modeled physics for stage separations and off-nominal trajectories as primary C++ developer in Unreal Engine 5.
- Earned a fully funded trip to NASA's Johnson Space Center; youngest invited speakers at the HELIX Discovery Showcase.

CubeSat, MIT Beaver Works Summer Institute, Co-Captain Dec 2024 – Aug 2025

- Placed 4th nationally in the BWSI Build-a-CubeSat Challenge with a wildfire-detection satellite that ran a vision language model trained on NASA GOES data.

FIRST Tech Challenge Team 24536, Mechanical Lead Oct 2023 – Apr 2025

- Designed competition mechanisms and integrated the full sensor suite, earning the Innovate Award.

Math League, Edison Academy Magnet School, Executive Board Aug 2023 – Jun 2026

- Secured sponsorships and authored problems for the MCAMC, raising \$3,000+ for the team's ARML trip.
- Served as Lead Coach at the Central Jersey Math Circle, the first middle school math circle in central New Jersey.

EDUCATION

Carnegie Mellon University

Expected May 2029
B.S. Mechanical Engineering + Robotics
Pittsburgh, PA

Current Coursework: Fundamentals of Mechanical Engineering, Concepts of Mathematics, Linear Algebra & Vector Calculus, Click Hack Rule (Cybersecurity), Putnam Seminar

Edison Academy Magnet School

Class of 2026
Mechanical/Civil Engineering Track ·
Weighted GPA 4.61 / 4.0
Edison, NJ

Middlesex College

Sep 2023 – Jun 2026
Dual Enrollment
Edison, NJ

Relevant Coursework: Statics, Linear Algebra, Differential Equations, Discrete Mathematics, Multivariable Calculus

AWARDS

USA Physics Olympiad Qualifier, Top 400 · 2025

NASA App Development Challenge Winner · 2025

4th Place Nationally, MIT BWSI Build-a-CubeSat Challenge · 2025

3rd Place, Rutgers McLaren Symposium · 2025

AIME Qualifier (2×), AMC 10B Top 5% · 2022, 2023

SKILLS

Programming

Python, C++, Java, MATLAB, LaTeX

Software

SolidWorks, Onshape, Unreal Engine 5, Blender, OVITO, AstrolmageJ

Fabrication & Electronics

CNC Machining, Laser Cutting, 3D Printing, Soldering, Circuit Design, Arduino, Raspberry Pi, Sensor Integration, Motor Control

Engineering & Research

Mechanism Design, Engineering Drawings, Photometry, Astrophotography, Statistical Data Analysis, Technical Writing